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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

DineStay: Restaurant, Hotel & Banquet Reservation Platform

Phase 1 – Minimum Viable Product (MVP)

Document Information

Document Title	DineStay Platform – Software Requirements Specification
Branding / Project Name	DineStay (Restaurants • Hotels • Experiences)
Version	1.0 (Phase 1 MVP Production Specification)
Prepared By	Future IT Touch Pvt. Ltd.
Prepared For	Client Executive Team / Stakeholders
Document Type	Software Requirements Specification (SRS)
UI Verification	35 High-Fidelity Mobile Application Wireframes & Web Portals
Status	Approved / Governing Implementation Document

1. Executive Summary

The **DineStay Platform** is an advanced digital hospitality and reservation ecosystem engineered to streamline table dining, hotel room reservations, and banquet/party hall event bookings into a single, cohesive experience. Built for both discerning consumers and hospitality business operators, DineStay bridges modern mobile convenience with centralized enterprise management.

The proposed Phase-1 MVP delivers end-to-end business capabilities: consumer onboarding, cross-vertical search and discovery (restaurants, hotels, and party halls in Chandigarh and surrounding regions), real-time table and room availability allocation, multi-channel digital payments (UPI, Cards, NetBanking, Wallets), dynamic discount coupons, and a comprehensive unified administration and partner portal suite.

The solution is architected as an interconnected ecosystem communicating via secure, rate-limited RESTful APIs built on Laravel, with an ultra-fast Flutter mobile application (Android and iOS) and responsive ReactJS portals.

2. Project Objectives

The primary strategic and technical objectives of Phase-1 include:

- **Digitize Restaurant Table Reservations:** Enable instant visual slot selection, party size specification, and guest special requests with instant SMS/Email confirmation.
- **Simplify Hotel Room Booking:** Real-time inventory check, room category comparison (King, Twin, Suite), multi-night stay rates, and check-in/out scheduling.
- **Banquet & Party Hall Inquiries:** Connect event organizers with premier party halls (50 to 500+ guest capacities) with venue specifications, catering, and valet details.
- **Instant Customer Discovery:** Geolocation-aware search across Chandigarh, filter chips for cuisines, star ratings, price tiers, and active promotional deals.
- **Automated Offers Engine:** Configurable coupon codes (e.g., DINE20, STAY15) with percentage/flat discounts and minimum spend thresholds.
- **Centralized Platform Administration:** Unified control over partner onboarding, verification of GST and licenses, commission accounting, and dispute handling.
- **Scalable Cloud Architecture:** Future-proof foundation supporting payments, loyalty rewards, multi-language support, and multi-city geographic rollout.

3. Project Deliverables

The DineStay Phase-1 MVP comprises six specialized software components:

Module	Target Platform	Core Technology	Scope Summary
Customer Mobile App	Android & iOS	Flutter 3.x / Dart	35 Verified UI Screens, Discovery, Bookings, Payments, Profile
Restaurant Business Portal	Web (Responsive)	ReactJS / Tailwind	Menu builder, slot allocation, live reservations table, analytics
Hotel Business Portal	Web (Responsive)	ReactJS / Tailwind	Room category management, daily tariff control, guest check-ins
Party Hall Business Portal	Web (Responsive)	ReactJS / Tailwind	Hall calendar, booking leads, package pricing, catering packages
Administration Portal	Web (Desktop)	ReactJS / AdminKit	Partner KYC approval, global coupons, revenue audits, user bans
Backend API Services	Cloud Server	Laravel 11 REST API	JWT authentication, booking state engine, notifications, cron jobs
Database Tier	Managed Server	MySQL 8.0 / Redis	Relational schema, transaction safety, Redis session & rate limiting

4. System Architecture & Topology

DineStay adopts a scalable client-server architecture. All frontend client applications communicate exclusively with the Laravel REST API layer over secure TLS 1.3 encrypted HTTPS channels.

Client Layer:	Flutter Mobile App (iOS / Android) • ReactJS Web Portals (Restaurant, Hotel, Party Hall, Admin)
Security & Gateway:	Cloudflare WAF • SSL/TLS 1.3 • Rate Limiting • CORS Protection • JWT Bearer Tokens
Application Layer:	Laravel 11 RESTful API Service • Controllers, Middleware, Form Requests, Observers
Integrations Layer:	Payment Gateway (UPI/Cards/NetBanking) • SMS Gateway (OTP) • AWS S3 / Cloudinary (Media)
Persistence Layer:	MySQL 8.0 (ACID Relational Storage) • Redis 7.x (Caching, Slot Locking, Queues)

5. User Roles & Access Control

DineStay implements rigorous Role-Based Access Control (RBAC) across five user categories:

5.1 Platform Administrator

The superuser with global oversight and configuration privileges across the entire DineStay ecosystem:

- **Business KYC & Onboarding:** Review submitted business licenses, GST certificates, and physical address proof for restaurants, hotels, and halls before public listing.
- **Account Moderation:** Instant suspension or restoration of offending partner accounts or abusive customer accounts.
- **Financial & Commission Auditing:** Set commission percentages per category, monitor gross transactional revenue, and generate payout sheets.
- **Global Campaign Engine:** Create platform-wide promotional codes (e.g., WELCOME50, FESTIVE) and manage hero app banners.

5.2 Restaurant Partner

- Manage restaurant profile, culinary cuisines, operating hours, and photo gallery.
- Configure visual digital menu with categorization, pricing, and Veg/Non-Veg badges.
- Set table capacity, available time slots (12:00 PM – 10:00 PM), and accept/reject reservations.
- Launch venue-specific coupons and track daily guest covers and revenue.

5.3 Hotel Partner

- Manage property description, star rating, address, amenities (Pool, Wi-Fi, Spa, Gym, Restaurant).
- Configure room inventories: King Room, Twin Room, Suite Room with occupancy rules and pricing.
- Process check-in and check-out logs, view booking vouchers, and handle cancellation requests.

5.4 Party & Banquet Hall Partner

- Manage hall capacity limits (e.g., 50–500 guests), event types (Weddings, Corporate, Birthday).
- Manage package pricing, parking arrangements, and in-house catering menu options.

5.5 Customer / End User

- Search, filter, compare, reserve dining tables, book hotel stays, and request banquet quotations.
- View booking history, cancel reservations with automated policy enforcement, save addresses, and earn coupon discounts.

6. Customer Mobile Application – UI Screen Inventory

The customer mobile application consists of **35 purpose-built screens** matching the verified UI wireframe specification. Each screen has been designed according to DineStay's forest green & warm terracotta visual identity:

#	Screen Name	Category / Row	Key Functional Components & Interactivity
1	**Splash Screen**	Row 1: Onboarding	Brand logo, 'DineStay' logotype, tagline 'Restaurants • Hotels • Experiences', culinary backdrop.
2	**Onboarding 1**	Row 1: Onboarding	Restaurants focus, circular hero illustration, pagination indicator, Skip & Next CTA.
3	**Onboarding 2**	Row 1: Onboarding	Hotel stays focus, cozy luxury bed illustration, pagination indicator, Skip & Next CTA.
4	**Onboarding 3**	Row 1: Onboarding	Best offers focus, digital discount vouchers illustration, 'Get Started' action.
5	**Login Screen**	Row 1: Auth	Email/Mobile input, Password, Forgot Password link, Google & Apple social OAuth buttons.
6	**Create Account**	Row 1: Auth	Customer / Business pill toggle, Full Name, Mobile, Email, Password, Terms agreement checkbox.
7	**OTP Verification**	Row 1: Auth	6-digit auto-advancing PIN inputs, resend countdown timer (00:25), Secure verification CTA.
8	**Home / Explore**	Row 1: Discovery	Location picker (Chandigarh), notification bell, category icons (4), Flat 20% banner, Popular cards.
9	**Search & Filters**	Row 1: Discovery	Search input, dynamic filter chips (Cuisine, Rating 4.0+, Offers, Sort), search result count.
10	**Restaurant Details**	Row 2: Dining	Hero gallery, star badge, distance, Call / Directions / Share / Save, Overview / Menu tabs.
11	**Restaurant Menu**	Row 2: Dining	Menu category chips (Recommended, Starters, Mains), Veg/Non-Veg indicators, pricing, Add button.
12	**Table Reservation**	Row 2: Dining	Horizontal date picker, lunch/dinner time slots, guest counter [- 2 +], special request field.
13	**Review Reservation**	Row 2: Dining	Booking summary card, chosen venue, date, time, party size, notes, Confirm Reservation CTA.
14	**Party Halls List**	Row 2: Banquets	Found halls count, venue photo cards (The Grand Hall, Royal Banquet), pricing from ■25,000.
15	**Party Hall Details**	Row 2: Banquets	Grand ballroom hero, tags (Weddings, Corporate), capacity (50-300), parking, in-house catering.
16	**Hotels Search**	Row 2: Hotels	Dates & occupancy bar, Price/Rating chips, Hotel cards (Hyatt Regency, Taj) with amenities.
17	**Hotel Details**	Row 2: Hotels	Luxury room hero, amenities grid (Pool, Wi-Fi, Spa, Dining), about property, View Rooms CTA.
18	**Select Room**	Row 3: Stays	Room cards: King Room (■9,800), Twin Room (■11,500), Suite Room (■15,200), Select button.
19	**Hotel Booking Review**	Row 3: Stays	Check-in/out dates, price computation (Room ■9,800 + Taxes ■1,200 = ■10,200), Proceed to Payment.
20	**Payment Method**	Row 3: Checkout	Payment options: UPI (GPay/PhonePe), Credit/Debit Card, Net Banking, Wallet, Save card option.

21	**Booking Confirmed**	Row 3: Checkout	Celebratory checkmark, Booking ID #BR25678, View Booking & Share actions.
22	**My Bookings**	Row 3: Account	Tabs: Upcoming (2), Past (5), Cancelled (1); booking cards with View Details & Cancel actions.
23	**Cancel Reservation**	Row 3: Account	Warning prompt, reservation summary, dropdown cancellation reasons, Confirm Cancel & Keep CTA.
24	**Notifications**	Row 3: Account	Filter tabs (All, Bookings, Offers, Updates), timestamped interactive notification cards.
25	**User Profile**	Row 3: Account	Avatar, name (Sneha Patel), contact details, links to Bookings, Addresses, Settings, Logout.
26	**Account Settings**	Row 3: Account	Profile info, change password, push notification toggles, language, privacy policy, app version.
27	**Offers & Coupons**	Row 4: Promotions	Coupon cards: FLAT 20% OFF (DINE20), FLAT 15% OFF (STAY15), category tabs, one-tap Apply.
28	**Favorites Wishlist**	Row 4: Discovery	Saved restaurants, hotels, and party halls with instant booking shortcuts.
29	**Saved Addresses**	Row 4: Account	Home & Work address cards (Sector 26 / Sector 22 Chandigarh), '+ Add New Address' action.
30	**Help & Support**	Row 4: Support	Interactive FAQ accordion, Contact Form, Report Issue, Feedback, and Direct Call shortcut.
31	**No Results Found**	Row 4: States	Clean empty search illustration, helpful corrective tips, 'Clear Filters' recovery button.
32	**Loading State (List)**	Row 4: States	CSS shimmer skeleton cards matching restaurant and hotel search result layouts.
33	**Loading State (Grid)**	Row 4: States	Skeleton placeholders for image thumbnails and detail cards during network fetch.
34	**Error State**	Row 4: States	'Something Went Wrong' illustration, descriptive error message, 'Try Again' reload CTA.
35	**Empty Bookings**	Row 4: States	Folder/calendar icon, 'No Bookings Yet' notice, 'Explore Now' navigation shortcut.

7. Detailed Functional Specifications

7.1 Authentication & Security Module

- **Dual User Registration:** Segregated signup workflows for Consumers and Business Partners with field validations.
- **OTP Phone Verification:** 6-digit cryptographic OTP generation with 5-minute expiry and 60-second resend cooldown.
- **Token Lifecycle:** JWT stateless tokens with 30-day refresh lifetime and instant revocation on logout/password change.

7.2 Search & Geolocation Discovery

- **Multi-Criteria Query Engine:** Full-text fuzzy search across business names, menu items, cuisines, and landmark sectors.
- **Filter Combination:** Intersecting filters: Cuisine (North Indian, Italian, Continental), Rating (4.0+), Price Tiers (■ to ■■■■), and Active Offers.

7.3 Dining Table Reservation Workflow

- **Dynamic Slot Allocation:** Real-time checking against restaurant table limits for requested time (12:00 PM – 10:00 PM).
- **Reservation State Machine:** Pending → Confirmed → Seated → Completed / Cancelled.
- **Special Accommodations:** User request capture (e.g., 'Window seat, Anniversary decor').

7.4 Hotel Stay & Room Selection Engine

- **Nightly Tariff Calculation:** Base room rate × Number of nights + 18% GST and municipal hospitality cess.
- **Room Inventory Locking:** 10-minute temporary checkout hold preventing double-booking during payment.

7.5 Payment Gateway Integration

- **Supported Modes:** Unified Payments Interface (UPI via GPay, PhonePe, Paytm), Credit/Debit Cards (3D Secure 2.0), Net Banking (40+ Indian banks), and Wallets.
- **Webhook Reconciliation:** Automated server-to-server callback processing ensuring 100% booking confirmation accuracy.

7.6 Promotional Discount Engine

- **Coupon Logic:** Validation of active date range, minimum cart amount, single-use per customer rules, and maximum discount caps.

8. Business Portals & Administration Console

In addition to the mobile app, the platform delivers three dedicated web management consoles:

8.1 Restaurant & Hotel Business Portal

- **Operational Dashboard:** Real-time KPI cards for Today's Bookings, Expected Covers, Active Room Occupancy, and Daily Revenue.
- **Booking Management:** Color-coded calendar timeline view. Ability to accept walk-ins, mark no-shows, and cancel with remarks.
- **Profile & Catalog Builder:** Upload high-res photography, update business hours, and toggle menu item out-of-stock.

8.2 Central Administration Portal

- **Partner Approval Pipeline:** Document viewer for FSSAI licenses, GSTIN validation, and bank passbooks before going live.
- **Commission & Payout Engine:** Configurable commission rates (e.g., 8% for dining, 12% for hotel bookings) with automated settlement logs.
- **Platform Audit & Security:** Immutable activity audit trail capturing every administrative modification.

9. Revenue Model

Revenue Stream	Mechanism	Target Economics
Transaction Commission	Percentage levied on confirmed hotel stays and dining vouchers.	8% – 15% per confirmed booking
Featured Placement	Priority ranking on explore screen and top category badges.	Monthly subscription (■3,000 – ■10,000 / mo)
Banquet Lead Generation	Fixed fee per qualified customer lead for high-ticket party halls.	■500 – ■1,500 per verified event lead

10. Non-Functional Quality Requirements

Dimension	Requirement Specification	Verification Metric
API Performance	Average response time under standard server load.	< 200ms latency (95th percentile)
Mobile Fluidity	Native 60 FPS transitions, skeleton shimmer during data load.	Startup time < 2.5s on mid-tier Android
Availability & Uptime	Redundant server deployment with automated failover.	99.5% Service Level Agreement (SLA)
Data Security	OWASP top-10 mitigation, SQL injection guard, Argon2id passwords.	Zero critical vulnerabilities on pen-test

11. Project Implementation Timeline

The Phase-1 development schedule spans **45 to 60 working days** structured into six milestones:

Phase	Duration	Core Deliverables
1. UI/UX & Wireframing	10 Working Days	Complete 35 screens design system, brand assets, interactive click-through.
2. Database & API Engine	20 Working Days	Laravel schema, JWT auth, business endpoints, booking state machine.
3. Mobile App Development	20 Working Days	Flutter frontend for 35 screens, state management, payment gateway integration.
4. Web Admin & Partner Portals	15 Working Days	ReactJS portals for business verification, reservations, and analytics.
5. QA & Security Audits	10 Working Days	End-to-end integration testing, boundary edge tests, performance optimization.
6. Staging & Production Launch	5 Working Days	Cloud server provisioning, CI/CD pipelines, Google Play & App Store submissions.

12. Commercial Proposal & Payment Milestones

Particulars	Scope Covered	Amount (INR)
Phase 1 MVP Development	Full Flutter Mobile App (35 Screens) + 3 Web Portals + Laravel Backend + Database + 3 Months Support	**■75,000**
Applicable GST	Standard Indian Goods & Services Tax (18% Extra)	■13,500
Total Investment	Comprehensive Phase 1 Turnkey Package	**■88,500**

Payment Schedule

Milestone	Trigger / Deliverable Stage	Percentage	Amount
Milestone 1	Project Initiation & Architecture Sign-Off	20%	■15,000
Milestone 2	UI/UX Wireframes & 35 Screens Prototype Approval	30%	■22,500
Milestone 3	Backend REST APIs & Portal Integration Completion	20%	■15,000
Milestone 4	UAT Sign-off, App Store Build Delivery & Launch	30%	■22,500

13. Warranty & Post-Launch Support

Future IT Touch Pvt. Ltd. provides **3 Months of Complimentary Technical Support** commencing from the date of final delivery. Support covers defect resolution, bug fixes, and server performance optimization. Feature expansions and UI redesigns outside the 35 verified screens will be scoped separately.

14. Acceptance & Sign-Off

This Software Requirements Specification serves as the binding contractual baseline for Phase-1 development of the DineStay Platform.

Prepared By:	**Approved By:**
Future IT Touch Private Limited Sector 34 A, Chandigarh Authorized Signatory: _____ Date: _____	**Client Stakeholders / DineStay Platform** Enterprise Client Authorized Signatory: _____ Date: _____